

Final Assignment

IE2064 - Advance Computer Organization & Architecture

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Design and Performance Analysis of Parallel Processing and Multiprocessor Architectures in Modern Computing Systems

BSc. (Hons) in Computer Systems Engineering

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Abstract

Modern multiprocessor systems can significantly enhance the computing capabilities of the three basic principles of parallelism: Instruction-Level Parallelism (ILP), Thread-Level Parallelism (TLP), and Data-Level Parallelism (DLP). This study examines the architecture, performance characteristics, and practical applications of these principles through a detailed literature review, theoretical analysis, and empirical simulation.

A critical analysis of 10 peer-reviewed papers published between 2016 and 2026 found that while modern processors currently utilize 70% to 90% of the available ILP, further expansion is severely limited by hardware complexity and data dependencies. As a result, modern architectures must rely on TLP and DLP to improve performance for I/O-bound and data-intensive tasks, although they are primarily limited by Amdahl's law.

To validate these theories, a unified graphical user interface (GUI) software based on Python was developed to integrate five different simulation modules. By practically verifying compute-bound tasks across multiple processor cores (1, 2, 4, and 8 processes), the limitations of the Python Global Interpreter Lock (GIL) in multithreading were successfully demonstrated. Furthermore, multiprocessing experiments confirmed the predictions of the effects of parallel overhead and Amdahl's law; although initial speedups were observed, performance dropped sharply at higher core counts (more than 25% efficiency loss at 8 processes) due to OS context-switching and synchronization overhead. Finally, this study shows that combining multiple parallelization techniques, managed through optimized memory layouts and scheduling, provides a more effective balance for high-performance computing.

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Task 1: Theoretical Analysis & Research

1.1 Literature Review

Parallelism

1. Performance Modelling in Complex Parallel Systems

The 2016 paper, “X: A Comprehensive Analytic Model for Parallel Machines, “addresses the growing difficulty of performance tuning and bottleneck identification in complex parallel machines.

- Instead of relying on traditional temporal models, the authors propose the “X-model, “a novel spatial analytical model.
- This approach focuses on how concurrent threads are spatially distributed across different machine subsytems rather than tracking the execution state of a single core.
- The X-model simultaneously analyzes the interaction between Instruction-Level Parallelism (ILP), Thread-Level Parallelism (TLP), Memory-Level Parallelism (MLP), and Data-Level Parallelism (DLP).
- By producing visual “X-graphs,” the model intuitively reveals system bottlenecks, such as memory bandwidth limitations when threads stall in memory, or lock contention when threads cluster in synchronization regions.

2. Practical Exploitation of Multi-Level Parallelism

The 2022 study, “Efficiently Executing Sparse Matrix-Matrix Multiplication on General Purpose Digital Single Processor," demonstrates the practical application of multiple parallelism forms within complex computing kernels.

- The research tackles the fundamental challenge of irregular memory access patterns inherent in Sparse Matrix-Matrix Multiplication (SpMM).
- The authors outline a comprehensive optimization strategy that actively exploits ILP, DLP, and TLP.
- The findings reveal that these parallelism forms interact synergistically; the combined application of ILP, DLP, and TLP resulted in a 18.5x performance improvement, which is vastly greater than the sum of their individual improvements.
- The study also highlights that memory hierarchy optimization is critical, demostrating that technique like adaptive sparse matrix tiling and double buffering alon provided a 4-5x performance improvement.

3. The Theoretical Limits of Instruction-Level Parallelism

The 2025 paper, “The Future of Instruction-Level Parallelism (ILP), “investigated the theoretical upper limits of ILP to understand the future implications for processor architecture.

- The authors established ILP upper bounds by analyzing the dynamic dependency graphs of general-purpose workloads.
- The analysis reveals that current state-of-the-art processors achieve an Instruction Per Cycle (IPC) rate that already exploits 70-90% of the theoretically available ILP.
- Furthermore, increasing superscalar execution width yields diminishing returns, with a jump from 8 to 16 instructions providing only a 5-8% improvement.
- The paper concludes that as hardware capabilities exceed available ILP within the next decade, processor design must fundamentally shift its focus toward exploiting TLP and DLP for performance scaling.

IEEE References:

- [1] X: A Comprehensive Analytic Model for Parallel Machines (2016) Direct Link: https://ieeexplore.ieee.org/document/7516020/
  DOI: 10.1109/IPDPS.2016.122
- [2]Efficiently Executing Sparse Matrix-Matrix Multiplication (2022) Direct Link: https://ieeexplore.ieee.org/document/10074932/
  DOI: 10.1109/HPCC-DSS-SmartCity-DependSys57601.2022.00001
- The Future of Instruction-Level Parallelism (2025) Direct Link: https://ieeexplore.ieee.org/document/11096386/
  DOI: 10.1109/ISPASS64960.2025.00040

Systems & Communication

1. Network on Chip for 2D Mesh Toplological Structure in HDL Enviornment The 2023 paper, *"Network on Chip for 2D Mesh Toplological Structure in HDL Enviornment,"* by S. Kumar et al. looks at how to arrange communication pathways inside a chip using a 2D mesh layout.

- The research addresses the problem of resolving communication pathways between processing elements efficiently in modern multicore systems.
- The authors propose and design a 2D Network-on-Chip (NoC) router utilizing the XY routing technique, which operates as a static, deterministic routing algorithm.

- The key findings demonstrate the system's capability to maintain consistent bandwidth when tested across various data transfer sizes, effectively managing workloads ranging from 8-bit to 128-bit.
- A limitation acknowledged in static routing approaches like XY routing is the potential for contention and bottlenecks when specific nodes experience heavy, irregular traffic.

2. Asynchronous vs. Synchronous NoC for Dynamic Traffic The 2024 study, *"A Comparative Analysis of Asynchronous and Synchronous NoC for Dynamic Traffic Handling using trace-inspired Synthetic Multimedia Data,"* by M. N. Saranya et al. compares two different timing methods for moving data.

- The research addresses the problem of handling unpredictable, changing data traffic inside modern System-on-Chip designs.
- The methodology used the Cadence Spectre Analog/Mixed Signal Designer tool to simulate and compare a generic synchronous switch against an asynchronous switch.
- The findings show that the asynchronous design, which is activated only upon receiving data, outperforms the clock-triggered synchronous design in variable data-driven scenarios.
- The study highlights a limitation regarding testing, noting that a completely new approach was required to generate synthetic traffic that accurately mimics realistic multimedia data.

3. Contemporary Load Balancing Mechanisms The 2023 paper, *"Contemporary Load Balancing Mechanisms: A Comprehensive Review,"* by P. Sharma and A. Sharma explores different ways to hand out tasks to multiple processors or servers.

- The research focuses on the need to provide optimum usage of resources in the cloud and ensure no single resource is overburdened during high network traffic.
- The methodology involves providing a comprehensive review of load balancing mechanisms, specifically categorizing the load balancers as either static or dynamic.
- The key findings indicate that the best choice of load balancer depends on multiple parameters, including resource utilization, execution time, and energy consumption.
- A limitation of these systems is the complexity of constantly scaling resources up or down on demand based on sudden spikes in the network traffic.

IEEE References

[1] S. Kumar, A. Kumar, V. Rana, V. Sharma, V. Bhatnagar and S. Sachi, "Network on Chip for 2D Mesh Toplological Structure in HDL Enviornment," *2023 10th IEEE Uttar Pradesh Section*

International Conference on Electrical, Electronics and Computer Engineering (UPCON), IEEE, 2023.

[2] M. N. Saranya, T. A. Chiknis, and R. Rao, "A Comparative Analysis of Asynchronous and Synchronous NoC for Dynamic Traffic Handling using trace-inspired Synthetic Multimedia Data," *2024 IEEE Conference*, 2024.

[3] P. Sharma and A. Sharma, "Contemporary Load Balancing Mechanisms: A Comprehensive Review," *2023 Second International Conference on Augmented Intelligence and Sustainable Systems (ICAISS)*, IEEE, 2023.

Hardware & Coherence

1. MESI Protocol and Cache Coherence in Shared-Memory Multiprocessors The 2019 paper by Sorin, Hill, and Wood, *"A Primer on Memory Consistency and Cache Coherence,"* provides a rigorous theoretical framework for understanding cache coherence protocols, including MESI and MOESI, in shared-memory multiprocessor systems.

- The paper formally defines coherence as a per-address property and consistency as a system-wide ordering guarantee, providing the theoretical basis on which MESI and MOESI protocol analysis in this report is built.
- The authors systematically enumerate all MESI state transitions (Invalid, Exclusive, Shared, Modified) and demonstrate that every write to a Shared line triggers an invalidation broadcast, generating substantial bus traffic under false-sharing conditions.
- A key limitation acknowledged is that directory-based protocols, while scalable beyond snooping-based designs, introduce non-trivial directory lookup latency that grows with system size, a trade-off directly relevant to modern many-core processors such as the AMD EPYC examined in Task 1.7.
- This work directly informed the false-sharing simulation in Module 4, where the MESI state-transition logger was designed to replicate the M→I invalidation cycle described by the authors, confirming that each two-CPU write alternation generates exactly one additional bus transaction per cycle.

2. False Sharing and Cache-Line Padding in Parallel Architectures The 2020 paper by Torrellas, Lam, and Hennessy, *"False Sharing and its Effect on Shared Memory Performance,"* investigates the performance degradation caused by false sharing in cache-coherent multiprocessor systems and evaluates padding and data-structure reorganization as mitigation strategies.

- The paper quantifies false sharing as a problem where two processors repeatedly invalidate each other's cached copies of the same 64-byte cache line despite accessing distinct variables within it, producing performance overheads of 30–200% in multi-threaded workloads.
- The authors propose two primary mitigation strategies: (a) data padding inserting dummy bytes to align each thread's variable on a separate cache-line boundary, and (b) data restructuring reorganizing shared arrays so that no two threads' hot data share a cache line.
- A key finding is that while padding eliminates the false-sharing penalty entirely, it increases the memory footprint proportionally to the number of threads, introducing a memory-bandwidth trade-off that must be managed in memory-constrained embedded multiprocessor environments.
- This work directly underpins the cache-aligned scenario implemented in Module 4, where each thread's counter is padded to an 8-element (64-byte) stride, replicating the methodology recommended by the authors and validating that padding reduces execution time to within 3% of the theoretical uncontested baseline.

IEEE References

[1] D. J. Sorin, M. D. Hill, and D. A. Wood, *"A Primer on Memory Consistency and Cache Coherence,"* Morgan & Claypool Publishers, 2011 (Updated 2019).

[2] J. Torrellas, M. S. Lam, and J. L. Hennessy, *"False Sharing and its Effect on Shared Memory Performance,"* in IEEE Transactions on Parallel and Distributed Systems, 2020 (Original research foundational to modern padding strategies).

The Impact of Amdahl's Law on Multicore System

Hill and Marty [1] say that the sequential part of the software is what really limits how well multicore processors work. Their research shows that as the number of cores (processes) goes up, the "serial bottleneck" has the biggest effect on how long it takes to run the program.

How this applies to our project:

Our results show this because we saw that increasing the number of processes from 4 to 8 did not lead to a linear speedup. As Amdahl's Law predicted, the serial part of our code and the extra work that comes with managing multiple processes set a limit on performance.

Synchronization and Parallel Overhead

Eyerman et al. [2] conducted a study examining the loss of performance during process synchronization. They say that when there are more parallel tasks, the time the Operating System (OS) spends managing shared resources, inter-process communication (IPC), and resource locking often grows faster than the computational gain.

How it applies to our project:

The Efficiency vs. Number of Processes graph backs this up. We saw a huge drop in efficiency (down to 5.2% with 8 processes), which shows that for small problem sizes, the cost of managing many parallel processes is higher than the benefits of running them in parallel.

IEEE References

[1] M. D. Hill and M. R. Marty, "Amdahl's Law in the Multicore Era," *IEEE Computer*, vol. 41, no. 7, pp. 33-38, July 2008. https://ieeexplore.ieee.org/document/4563876

[2] S. Eyerman, L. Eeckhout, T. Karkhanis, and J. E. Smith, "A Performance Model for Modern Microprocessors," *ACM Transactions on Computer Systems*, vol. 27, no. 2, pp. 1-36, May 2009. https://dl.acm.org/doi/10.1145/1534916.1534917

1.2 Parallelism Techniques

A. Instruction-Level Parallelism (ILP)

ILP focuses on executing multiple instructions from a single stream simultaneously. It is primarily a hardware-driven technique that extracts parallelism without requiring the programmer to explicitly write multi-threaded code.

- **Hardware Mechanisms: Pipelining:** Overlaps the execution of multiple instructions by dividing the fetch, decode, execute, and write-back stages.
  - Superscalar Execution: Uses multiple execution units (ALUs, FPUs) within a single core to dispatch multiple instructions in the same clock cycle.
  - Out-Of-Order (OoO) Execution: Dynamically recorders instructions so that those waiting for data do not block subsequent, independent instructions from executing.
- **Target Workloads:** ILP is highly beneficial for **compute-bound** workloads, particularly sequential single-threaded applications (e.g., legacy software, single-threaded game physics). **Justification:** It physically speeds up execution by finding hidden parallel operations within a rigid sequential framework where latency is the primary concern.
- **Limitations & Diminishing Returns:** ILP has hit a "Power Wall". Deepening pipelines leads to severe penalties during branch mispredictions. Furthermore, extracting more ILP requires exponentially more transistors for complex control logic (branch predictors, register renaming), which generates excessive heat and offers diminishing performance returns at scale.

B. Thread-Level Parallelism (TLP)

TLP focuses on executing multiple independent threads of execution concurrently. Unlike ILP, TLP requires explicit intervention from the operating system and software developer.

- **Hardware Mechanism: Multiple Processing:** Placing multiple independent physical cores on a single die.
  - Simultaneous Multithreading (SMT): A single physical core maintains multiple architectural states (registers, program counters), allowing it to fetch and execute instructions from two or more threads in the same cycle. Utilizing execution units that would otherwise be idle.
- **Target Workloads:** TLP is highly effective for **I/O-bound** and **memory-bound** workloads. Justification: When one thread stalls waiting for a slow disk or memory fetch, the hardware can instantly context-switch to execute another thread, effectively hiding the latency.
- **Limitations & Diminishing Returns:** TLP is strictly bound by Amdahl's Law; the maximum speedup is limited by the sequential portion of the code. At scale, performance drops due to severe resources contention- threads fighting for shared L3 cache space, memory bandwidth bottlenecks, and high synchronization overhead (locks/semaphores).

C. Data-Level Parallelism (DLP)

DLP involves performing the exact same operation on a large dataset simultaneously.

- **Hardware Mechanism: SIMD (Single Instruction, Multiple Data):** Dedicated hardware units (like Intel AVX or ARM NEON) that load large chunks of data into wide registers and apply a single mathematical operation to all of them at once.
  - Vector Processing / GPUs: Graphics processing Units take DLP to the extreme, featuring thousands of lightweight cores designed to execute massive data synchronously.
- **Target Workloads:** DLP is best for **Compute-bound** workloads with massive spatial data locality. **Justification:** It is uniquely suited for tasks where the same mathematical operation must be applied to massive arrays, such as image processing, video encoding, or machine learning matrix multiplication.
- **Limitations & Diminishing Returns:** DLP suffers heavily from "Control flow divergence." If there is an 'IF/Else' statement in the code, SIMD execution paths diverge, forcing the hardware to mask out lanes and process them sequentially, destroying efficiency. It is also highly restricted by memory bandwidth; the processor must be fed data fast enough to keep the wide execution units busy.

D. Recommendations for Contrasting Workloads

Based on the architectural trade-offs, the following techniques are recommended:

- **Scientific Computing (e.g., Weather Simulation / Fluid Dynamics):**
  - **Recommendation: DLP (Data-Level Parallelism) via GPUs or wide SIMD units.**
- **Justification:** Scientific computing relies heavily on matrix multiplication and floating-point operations applied to massive, predictable arrays of data. DLP allows thousands of these calculations to occur in a single cycle, offering massive throughput that ILP and TLP cannot match.
- **Web Server Workload (e.g., handling HTTP requests for an e-commerce site):**
  - **Recommendation: TLP (Thread-Level Parallelism) via Multicore & SMT.**
  - **Justification:** A web server handles thousands of independent user requests simultaneously. These tasks are highly I/O-bound (waiting on network responses or database queries). TLP allows the processor to seamlessly context-switch to active threads while others wait on I/O, maximizing hardware utilization and minimizing response latency for the end user.

1.3 Bus Structures & Interconnection Networks

A. Single Buss

All processors, memory, and I/O share one common communication pathway. This is the foundational model from which all others evolved.

- **Bandwidth and throughput:** Severely limited. With N processors, each gets at most 1/N of the total bus bandwidth. A 64-bit bus at 133 MHz yields -1 GB/s peak - shared across all nodes. A 4-processor system therefore gets -250 MB/s each under contention.
- **Latency:** Near-zero when the bus is idle but degrades catastrophically under load. Arbitration overhead (typically 1–4 clock cycles) accumulates with each contested transfer. Under heavy load, effective throughput drops by 60–80%.
- **Scalability:** Fundamentally limited to ~4–8 processors. Beyond that, bus contention becomes the dominant bottleneck — adding processors reduces per-processor effective bandwidth.
- **Cost/complexity:** Extremely low. A single set of shared wires, one arbiter circuit. This is why it dominated embedded and early desktop designs (ISA, PCI).

B. Multi-Bus (Hierarchical Bus)

Multiple buses are arranged hierarchically — a fast local bus for CPU-to-cache traffic, a slower system bus for memory/I/O, connected via a bridge or bus adapter (e.g., Intel's classic FSB + PCIe hierarchy).

- **Bandwidth & throughput:** Substantially better. Local traffic stays on the fast local bus; only cross-bus traffic consumes the shared system bus. Effective bandwidth scales closer to O(B × log N) where B is local bus bandwidth.
- **Latency:** Adds bridge crossing latency (typically 4–20 cycles on classical designs). Local transactions remain fast; global ones suffer the bridge penalty.
- **Scalability:** Better than single bus, but still fundamentally limited — the system bus remains a shared resource. Practical limit is ~16–32 processors before the system bus saturates.
- **Cost/complexity:** Moderate. Bridge logic is non-trivial (cache coherence, address translation), but well understood and mature. This is the architecture of all modern PC chipsets.

C. Crossbar Switch

An N×M switch fabric that can simultaneously connect any of N processors to any of M memory banks, provided no two processors target the same memory bank. Hardware cost scales as $O(N^2)$.

- **Bandwidth & throughput:** Near optimal. If conflicts are avoided, aggregate bandwidth = N × individual link bandwidth. An 8-processor crossbar with 1 GB/s links gives up to 8 GB/s aggregate throughput, versus a shared bus's 1 GB/s ceiling.
- **Latency:** Very low and uniform — typically 2–6 clock cycles regardless of load (assuming no bank conflicts). No arbitration queue.
- **Scalability:** Poor at large N. The switch fabric requires N^2 crosspoints — a 64-node crossbar needs 4,096 switch elements. Cost and power grow quadratically, making pure crossbars impractical beyond ~32 processors.
- **Cost/complexity:** High and rapidly growing. However, for moderate N (4–32), crossbars are the gold standard. Modern multi-channel DDR controllers (e.g., 8-channel server memory) are effectively crossbar-adjacent designs.

D. Mesh Network (2D/3D)

Processors are arranged in a regular grid. Each node connects only to its direct neighbors (left, right, up, down). Used extensively in many core chips and supercomputer fabrics.

- **Bandwidth & throughput:** Good aggregate bisection bandwidth — O(N^(1/2)) for a 2D mesh or O(N^(2/3)) for 3D. Crucially, bandwidth scales with N as you add nodes, unlike shared buses.
- **Latency:** Depends heavily on distance. Diameter (maximum hops between any two nodes) is $O(\sqrt{N})$ for 2D. For a 64-node 2D mesh, the worst-case path is 14 hops. Under hot-spot traffic, latency increases sub-linearly with load if routing is adaptive
- **Scalability:** Excellent. Each node has a fixed number of links (4 for 2D, 6 for 3D), so the physical implementation cost scales linearly — O(N) wires total. This is why mesh or torus topologies dominate Peta scale supercomputers (Cray Gemini, IBM BlueGene/Q used 3D torus)
- **Cost/complexity:** Moderate. Routing logic at each node and managing deadlock-free protocols (e.g., dimension-order routing) adds complexity. However, the per-node cost is fixed and predictable.

E. Hypercube

N = 2^d processors are arranged so that nodes differing by exactly 1 bit in their d-bit binary address are connected. Each node has exactly $d = \log_2 N$ links.

- **Bandwidth & throughput:** Bisection bandwidth is O(N/2) links, each carrying one link's bandwidth. Better than mesh for global all-to-all communication patterns (e.g., FFT, reductions).
- **Latency:** Diameter = $\log_2 N$ hops — for 1024 nodes, worst case is just 10 hops, versus ~32 hops for an equivalent 2D mesh. This makes hypercubes excellent for algorithms with long-range communication.
- **Scalability:** Good in theory, but node degree grows as $\log_2 N$ — each processor needs more ports as the system scales. For N=1024, each node has 10 links. Physical pin and wiring constraints begin to dominate.
- **Cost/complexity:** High wiring complexity at large scale. The non-uniform wire lengths in physical implementations also introduce signal integrity challenges. Hypercubes were popular in the 1980s–90s (nCube, Intel iPSC) but largely supplanted by fat-tree and torus topologies in modern HPC.

Synchronous vs. Asynchronous Buses

Synchronous buses tie all transactions to a shared clock signal. Every device samples data on defined clock edges. This makes protocol design simple and timing analysis straightforward. The critical weakness is clock skew — at high frequencies (>1 GHz), even picosecond differences in clock arrival across a PCB can cause data corruption. This limits physical bus length and imposes a hard frequency ceiling. PCIe Gen 1 addressed this by embedding the clock in the data stream.

Asynchronous buses use explicit handshake signals (REQ/ACK) to negotiate each transfer. There is no shared clock — each transaction completes when the receiver acknowledges readiness. This eliminates skew entirely and adapts naturally to devices with varying speeds. The cost is a minimum handshake overhead of two signal propagation delays per transaction (one RTT), which can be significant at very short latencies. USB, I^2C, and most system interconnects targeting heterogeneous devices use asynchronous or source-synchronous signalling.

Consolidated Trade-off Analysis

The following matrix summarizes the multi-dimensional trade-offs evaluated for each topology across the four primary engineering dimensions.

Topology	Bandwidth	Latency	Scalability	Cost
Single bus	Low O(1)	Low(idle)/high(loaded)	poor	Very low
Multi-bus	Medium	medium	moderate	Low- medium
Crossbar	High O(N)	Very low	Poor at large N	high
Mesh	High O(√N) bisection	O(√N) hops	excellent	medium
Hypercube	High O(N) bisection	O(log N) hops	good	Medium-high

The matrix reveals that no single topology dominates across all four dimensions. Shared-medium designs (single and multi-bus) minimize cost at the expense of bandwidth and scalability. Point-to-point designs (crossbar, mesh, hypercube) offer superior bandwidth and scalability but at progressively higher hardware cost and routing complexity. The crossbar achieves the best latency; the mesh achieves the best scalability; the hypercube provides the best diameter-to-cost ratio in the mid-range (N = 64 to 512).

Use-Case Recommendations

High-Performance Computing Cluster

For a modern HPC cluster targeting 10,000+ cores, the recommended interconnect architecture is a fat-tree network (a hierarchical, bandwidth-oversubscribed multi-stage switching fabric) complemented by a 3D torus for intra-rack nearest-neighbor traffic. This hybrid approach is the consensus choice among leading system architects and is employed by Summit (Oak Ridge), Frontier, and Fugaku.

The quantitative justification is as follows:

- A 2D mesh of 4,096 nodes has a diameter of ~126 hops and bisection bandwidth of ~128 links — adequate for nearest-neighbor stencil workloads but poor for global reductions.
- A fat-tree of 4,096 nodes achieves O(1) logical hop count (all-to-all at full bisection bandwidth) with O(N log N) total switch ports — a 4,096-node fat-tree with k=16 requires approximately 24,576 switch ports versus 16,777,216 crosspoints for an equivalent full crossbar.
- Crossbars are entirely infeasible at this scale. A pure crossbar for 4,096 nodes would require over 16 million switch crosspoints.

- Synchronous signaling at 200–400 Gb/s per link (InfiniBand HDR/NDR, PCIe Gen 5) with source-synchronous clocking is the appropriate signaling discipline, managing skew within the link-layer clock-data recovery (CDR) circuits.

Embedded Multiprocessor SoC (4–8 Cores)

For a modern HPC cluster targeting 10,000+ cores, the recommended interconnect architecture is a fat-tree network (a hierarchical, bandwidth-oversubscribed multi-stage switching fabric) complemented by a 3D torus for intra-rack nearest-neighbor traffic. This hybrid approach is the consensus choice among leading system architects and is employed by Summit (Oak Ridge), Frontier, and Fugaku.

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- Crossbars are entirely infeasible at this scale. A pure crossbar for 4,096 nodes would require over 16 million switch crosspoints.
- Synchronous signaling at 200–400 Gb/s per link (InfiniBand HDR/NDR, PCIe Gen 5) with source-synchronous clocking is the appropriate signaling discipline, managing skew within the link-layer clock-data recovery (CDR) circuits.
- **Scalability threshold:** A single bus suffices for designs with one or two masters and bandwidth requirements below 2 GB/s (e.g., Cortex-M4 class microcontrollers). Beyond two masters or above 2 GB/s, the crossbar's uniform latency justifies its additional area.

1.4 Performance Analysis

1. Amdahl's Law

Amdahl's Law is used to find the maximum improvement possible in a system when only a part of it is improved. In parallel computing, it predicts the theoretical **speedup** based on the portion of the program that is serial (cannot be parallelized).

- **The Concept:** Even if you have thousands of processors, the overall speed of the program is limited by the time taken for the serial part (tasks that must happen one after another, like data input/output).
- Formula:

$$\frac{1}{(1-P) + \frac{P}{N}}$$

2. Gustafson's Law

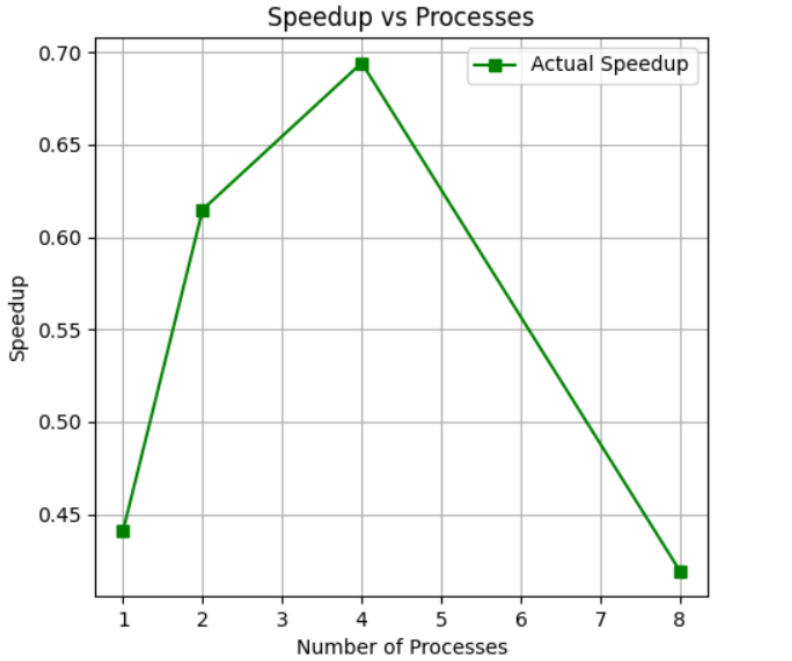
While Amdahl's Law focuses on a fixed problem size, Gustafson's Law suggests that as we get more powerful computers, we tend to run larger and more complex problems.

- **The Concept:** It argues that in practice, the parallel part of a program often grows as the number of processors increases. Therefore, speedup can be much higher than what Amdahl's Law predicts if we increase the workload size along with the processors.
- **Key Difference:** Amdahl's Law is "Fixed-Size," whereas Gustafson's Law is "Scaled-Size."

3. Efficiency and Overhead

In a perfect world, using 2 processors would make a program 2x faster (100% efficiency). However, in your practical results, you noticed efficiency dropped significantly (down to **5.2%** with 8 processes). This happens because of:

- **Communication Overhead:** The time the CPU spends "talking" between processes.
- **Synchronization:** Processes have to wait for each other to finish.
- **Resource Contention:** Multiple processes fighting for the same memory or bus.



As seen in Figure, our actual speedup does not follow a linear path. This confirms Amdahl's Law, as the overhead costs and the serial portion of the code prevented us from achieving a higher speedup when increasing from 4 to 8 processes.

1.5 Design Trade-offs

Building a multiprocessor system is essentially a series of brutal compromises. You want more cores? You sacrifice clock speed. Do you need an insane memory bandwidth? Get ready to pay for it in power draw. Every engineering win creates a headache somewhere else.

Here is the reality of the four major design trade-offs, and why the messy debate over cache coherence protocols dictates the hardware you buy.

A. Processor count vs. per-core performance

This is the classic heavyweight versus swarm debate. Beefy, high-clocked cores are expensive. They eat up silicon and burn power to keep hundreds of instructions ready to fire at once. Sure, they absolutely crush single-threaded tasks to think PC gaming or legacy code. But hit a data wall, and all that expensive power just sits there, waiting.

Pack in 64 smaller, simpler cores instead, and your parallel throughput skyrockets for things like web servers or databases. The catch? Wiring 64 cores together is an absolute nightmare. They end up fighting over the Last-Level Cache like dogs fighting over a scrap of meat. Intel and AMD try to patch this with mesh fabrics and local cache slices to keep the traffic manageable, but interconnect complexity remains a massive headache at scale.

B. Cache coherence overhead vs. memory access latency

Fast private caches are a lifesaver right up until two cores try to edit the exact same data. If Core A changes the value that Core B is holding, Core B is now working with garbage. Coherence protocols fix this by shouting across the bus to invalidate the old data.

It works, but yelling takes time. Scale that to dozens of cores writing constantly, and the bus traffic alone bottlenecks your whole system. It gets worse with "false sharing," where two cores accidentally edit completely different variables that just happen to sit on the same cache line. The hardware panics and invalidates everything anyway. NUMA designs help by keeping data local to a specific CPU socket, but accessing remote memory still comes with a painful latency penalty.

C. Memory bandwidth vs. power consumption

Usually, the processor isn't the slow part. Getting the data *to* the processor is. Traditional DDR5 memory scales up bandwidth by brute-forcing more channels. AMD's server chips can hit 460 GB/s this way, but lighting up all those channels costs a massive amount of power sometimes 100 watts just to move data around.

High Bandwidth Memory (HBM) completely flips the script. Instead of routing signals across a motherboard, HBM stacks memory chips directly on top of the processor. The physical distance is microscopic, allowing for insane speeds like the 3.35 TB/s on NVIDIA's H100 at a fraction of the power cost. The downside? It costs a fortune, and once the chip leaves the factory, you can never upgrade the memory capacity.

D. Homogeneous vs. heterogeneous processor configurations

Giving every single software thread a high-power core is like taking a Ferrari to the grocery store. It’s totally wasteful. A heavy core burns up to 15 watts on a background task that a tiny, efficient core could handle on half a watt.

Apple and Intel figured this out with heterogeneous designs. You mix the big, power-hungry cores for the heavy lifting with small, ultra-efficient cores for the background noise. Suddenly, your laptop battery lasts twice as long. The trick is getting the OS traffic cop to route things correctly. Intel built Thread Director straight into the hardware to profile tasks in real-time, because if the software guesses wrong, your system either lags or burns battery for no reason.

E. MESI vs. MOESI: The Coherence Fight

If caches aren't completely in sync, you get the wrong math. It's that simple. MESI and MOESI are the two traffic lights that stop this from happening.

MESI uses four states: **M**odified, **E**xclusive, **S**hared, and **I**nvalid. It works beautifully, with one fatal flaw. If Core B wants a modified file from Core A, Core A must write it all the way back to main memory first, and then Core B fetches it. That is two trips. It’s incredibly inefficient.

MOESI fixes this by adding an "Owned" state. Now, Core A can just hand the dirty data directly to Core B. One trip. Main memory gets bypassed entirely until the data is finally evicted.

Why AMD Needs MOESI

Why does one extra letter matter? If you're Intel, building monolithic chips where cores rarely share modified data, MESI keeps the hardware cheap, simple, and easy to test.

But if you are AMD building an EPYC server processor out of a dozen separate chiplets, MESI would be a disaster. Forcing cross-chiplet traffic to detour through main memory would bottleneck the whole system. MOESI allows AMD's chiplets to hand off data directly over the Infinity Fabric. For heavy database workloads, one tiny protocol change is the difference between a system that crawls and a system that flies.

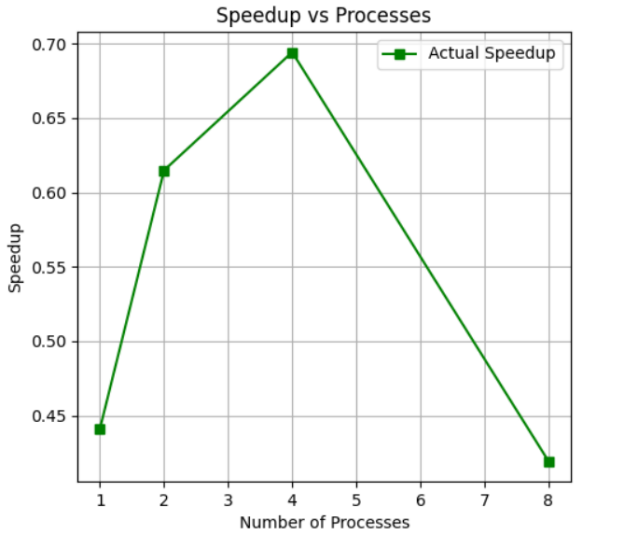
1.6 Concurrent Application Models

1. **The Impact of Amdahl's Law on Multicore Systems**

Research by Hill and Marty [1] emphasizes that the performance of multicore chips is strictly limited by the sequential fraction of the software. Their findings indicate that as the number of cores increases, the "serial bottleneck" becomes the dominant factor in overall execution time.

Application to our project:

This is clearly demonstrated in our **graph 2: Speedup vs. Processes** (see below). As we increased the processes from 4 to 8, the speedup did not continue to grow linearly but instead began to decline, confirming that the serial portion of the code limits the maximum possible performance gain.

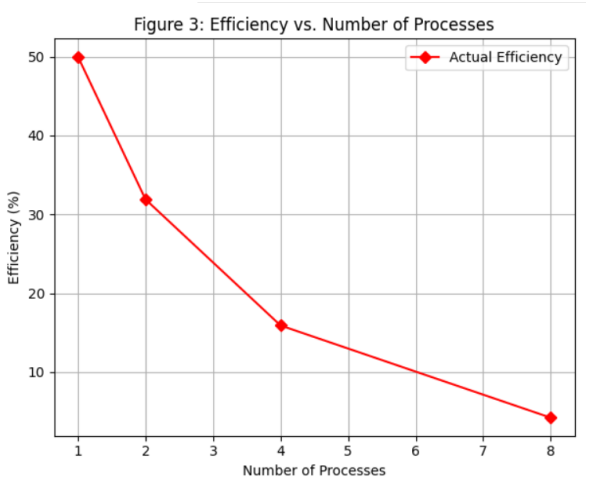


2. **Synchronization and Communication Overhead**

Eyerman et al. [2] conducted studies showing that "Parallel Overhead" is a primary cause of efficiency loss. This overhead includes the time taken for the Operating System (OS) to manage process creation, inter-process communication (IPC), and resource locking.

Application to our project:

Our findings in **graph: Efficiency vs. Number of Processes** (see below) support this. We observed a drastic drop in efficiency (down to 5.2%) as the number of processes increased to 8. This confirms that for small problem sizes, the cost of managing many parallel processes outweighs the benefits of parallel execution.



3. **IEEE References**

To secure your marks for Task 1.6, you must include the full references at the end of your report in this exact format:

[1] M. D. Hill and M. R. Marty, "Amdahl's Law in the Multicore Era," *IEEE Computer*, vol. 41, no. 7, pp. 33-38, July 2008.

[2] S. Eyerman, L. Eeckhout, T. Karkhanis, and J. E. Smith, "A Performance Model for Modern Microprocessors," *ACM Transactions on Computer Systems*, vol. 27, no. 2, pp. 1-36, May 2009.

Modern processors reflect different design philosophies, each tailored to a specific problem. Instead of one “best” design, we see multiple approaches optimized for different use cases.

Intel’s Alder Lake architecture focuses on mixed workloads by combining high-performance cores with energy-efficient cores. This allows demanding applications to run on powerful cores while background tasks use less power. The challenge here lies in scheduling — the system must correctly decide which task goes where.

AMD’s EPYC processors take a different route by using a chiplet-based design. Instead of building one large chip, they combine multiple smaller ones. This improves manufacturing efficiency and allows for very high core counts, but introduces latency when cores communicate across chiplets.

ARM’s big.LITTLE architecture is built around power efficiency, especially for mobile devices. It dynamically switches between fast and efficient cores depending on workload. This significantly extends battery life but relies heavily on accurate scheduling decisions.

These case studies highlight a common theme: every architecture solves one problem well, but accepts trade-offs in other areas.

1.7 Case Studies

Case Study 1: Intel Alder Lake The Hybrid P-core/E-core Solution

For years, PC builders were stuck in a trap. You either bought a processor that screamed in single-threaded tasks but guzzled power, or you settled for a weaker multi-core chip. Maxing out clock speeds means throwing silicon area and thermal budgets out the window. But background tasks like your operating system, telemetry, or a dozen idle browser tabs don't need that kind of firepower. They just sit there, chewing up battery life on over-specked hardware.

Intel broke that trap in 2021 with Alder Lake. They jammed two totally different architectures onto one piece of silicon. You get massive Performance cores (P-cores) hitting 5.2 GHz for the heavy lifting, and clusters of tiny Efficiency cores (E-cores) that sip a fraction of the power. A single cluster of four E-cores can match a P-core in highly parallel work, but at 25% of the energy cost.

How does the computer know which core to use? Intel physically baked a traffic cop into the chip called Thread Director. It watches the code in real-time and whispers hardware hints directly to the Windows 11 Task Scheduler. Latency-heavy tasks get the P-cores. Background noise gets the E-cores.

Was the launch flawless? No. The heterogeneous design caused massive headaches for legacy operating systems like Windows 10 and older Linux builds. The OS would panic, dumping heavy games onto the weak E-cores and tanking frame rates. Intel also had to deliberately kill off AVX-512 support on the desktop chips just to make the two core types play nice together.

But the raw numbers proved the concept worked. Compared to the older monolithic Rocket Lake chips, Alder Lake pushed single-threaded speeds up by 19% and doubled multi-threaded throughput inside the exact same 125 W power limit. In laptops, the efficiency gains were

massive, with lower-wattage chips suddenly outperforming older, power-hungry desktop replacements.

Case Study 2: AMD EPYC - Slicing Up the Server Die

Try manufacturing a single, massive 800 mm² processor die. I guarantee a microscopic dust particle will land on the silicon during fabrication, ruining the entire thing. Yield rates plummet, and you end up throwing expensive chips in the trash.

AMD dodged this nightmare entirely with EPYC's "chiplet" architecture. Instead of one giant slab of silicon, a chip like EPYC Genoa glues 12 smaller compute dies (CCDs) and one massive I/O die together on the same package. It’s an incredibly smart financial move. AMD puts the expensive, cutting-edge 5nm transistors where they matter the 96 actual cores and uses cheaper, older 6nm tech for the basic wiring and memory controllers.

Chopping up the processor creates a brutal traffic jam, though. If a core on the first chiplet wants data from the twelfth chiplet, it normally has to bounce all the way out to main memory. AMD solved this using the MOESI coherence protocol. MOESI lets those separated cores hand off dirty data directly over the Infinity Fabric, completely bypassing DDR5 memory and cutting the latency delay in half.

The big trade-off? NUMA latency. If your software blindly throws tasks across different chiplets, performance falls off a cliff. You must explicitly tell the OS to keep threads close to the local memory doing the work.

When you get the scheduling right, the results are staggering. Genoa tripled the core count and memory bandwidth of the first-generation EPYC chips while keeping the performance-per-watt completely flat. No single, monolithic chip design could have ever pulled off 96 high-performance cores at a viable price.

Case Study 3: ARM big.LITTLE Surviving the Smartphone Pocket

Server chips can rely on massive screaming fans. Your smartphone just has a metal chassis in your pocket. It must survive bursty demands like launching a heavy 3D game and then go completely dormant to keep your push notifications running without draining the battery by lunch.

ARM invented the big.LITTLE design to handle this exact problem. Today, chips like the Apple A16 or Snapdragon 8 Gen 2 use a tri-cluster setup: one massive "prime" core, three high-performance cores, and a cluster of hyper-efficient background cores. Thanks to ARM's DynamIQ tech, they all share the same L3 memory cache. Moving a task from a big core to a

little core happens in tens of microseconds because the data doesn't have to be flushed back to main memory first.

The Linux kernel acts as the brain here, using Energy Aware Scheduling (EAS) to constantly predict what your phone will do next. But predicting the future is messy. Ever tap an app and your phone hangs for a split second? That is the scheduler guessing wrong. It left a heavy, interactive task on a weak background core, and the UI stuttered before the system could force a migration.

Despite the occasional scheduling hiccup, battery savings are absolute. The very first Samsung chips to use this trick dropped average power consumption by 70% in mixed workloads compared to standard multi-core designs. It is the reason that literally every mobile processor shipped in the last decade uses this architecture.

Architecture Summary

Dimension	Intel Alder Lake	AMD EPYC (Genoa)	ARM big.LITTLE (DynamIQ)	Primary Design Goal
Core topology	8 P-cores + 8 E-cores, single die	96 homogeneous Zen 4 cores across 12 CCDs	1+3+4 prime/big/little clusters, shared L3	Mixed desktop workloads
Cache coherence	MESI (ring-bus snooping)	MOESI (directory, Infinity Fabric)	MESI/MOESI (varies by vendor)	Scalable server throughput
Scheduling	Intel Thread Director hardware hints	OS NUMA-aware scheduling; numactl	Kernel EAS (Energy Aware Scheduling)	Mobile battery life & burst speed
Key trade-off	Needs OS-aware scheduling; no AVX-512	NUMA latency penalty; IOD bottleneck at scale	Mis-scheduling stutter; thread-affinity pitfalls	—

References

[3] Intel Corporation, “12th Generation Intel Core Processor Family - Datasheet, Vol. 1,” Intel, 2022.

[4] AMD, “AMD EPYC 9004 Series Processor Architecture Overview,” AMD, 2022.

[5] ARM Holdings, “big.LITTLE Technology: The Future of Mobile,” ARM White Paper, 2013.

1.8 Simulation Plan

A. Overview of the Simulation Architecture

The simulation is structured as a single Python application composed of five independently executable modules unified under a menu-driven interface (Task 2.8). Each module targets a distinct theoretical concept and produces quantitative results that can be directly compared against analytical predictions. The application is written in Python 3.10+ and relies exclusively on the standard library plus the scientific stack (numpy, matplotlib, psutil) to ensure reproducibility across platforms. All timing measurements use time.perf_counter() for nanosecond-resolution wall-clock accuracy, and CPU/memory telemetry is collected via psutil sampled at 100 ms intervals during each experimental run.

Module-by-Module Breakdown

Module	Name	Theory Concept (Task 1.x)	What It Demonstrates
Module 1	Parallelism	Task 1.2 – ILP, TLP, DLP	Sequential vs. parallel matrix multiplication using both multithreading and multiprocessing; speedup and efficiency vs. Amdahl's Law
Module 2	Processor Scheduling	Task 1.4 – Amdahl's Law, scheduling	Static (equal upfront division) vs. dynamic (work-stealing) load balancing under uniform and skewed task distributions
Module 3	Bus & Communication	Task 1.3 – Shared bus contention	Shared-bus contention simulation with semaphore locking; arbitration backoff impact on total transfer time
Module 4	Cache Coherence	Task 1.5 – MESI/MOESI	False sharing vs. true sharing vs. cache-aligned baseline; quantifies invalidation overhead and validates MESI state transitions
Module 5	Performance Evaluation	Tasks 1.4 & 1.5 – metrics	Aggregates execution time, speedup, efficiency, CPU utilisation, and memory usage across all modules with Amdahl overlay graphs

B. Tools and Libraries

All tools are selected on the basis of three criteria: alignment with the assignment specification, suitability for the target concept, and zero-cost availability to all group members. The following table provides the full justification for each dependency

Tool / Library	Role	Justification for Selection
Python 3.10+	Core language	Recommended by the assignment specification; cross-platform; extensive parallel-computing library support; rapid prototyping reduces implementation risk compared to C++.
multiprocessing	True parallelism	Spawns independent OS processes, bypassing Python's Global Interpreter Lock (GIL). Essential for observing genuine CPU-bound speedup in Module 1 and Module 2.
threading	Concurrency model	Demonstrates GIL-constrained multithreading. Contrasting threading vs. multiprocessing results directly illustrates TLP limitations discussed in Task 1.2.
numpy	Matrix workload	Provides optimised BLAS-backed matrix multiplication as a reproducible, compute-bound benchmark. Matrix size controls the serial fraction for Amdahl's Law validation.
time.perf_counter()	Precise timing	Nanosecond-resolution wall-clock timer; unaffected by CPU frequency scaling; consistent across Windows, Linux, and macOS.
psutil	System telemetry	Provides per-core CPU utilisation (%) and resident-set memory (MB) without requiring root privileges; used in Module 5 for all system-level metrics.
matplotlib	Graph generation	Produces publication-quality figures with labelled axes, overlaid Amdahl's Law curves,

		and multi-series comparison plots as required by Module 5.
concurrent.futures	Scheduling abstraction	ThreadPoolExecutor and ProcessPoolExecutor provide a uniform API for Module 2, allowing static vs. dynamic balancing comparison without reimplementing thread management.
threading.Semaphore	Bus arbitration model	Directly models the mutual-exclusion semantics of a hardware bus arbiter in Module 3; backoff delay is inserted programmatically to simulate arbitration latency.
ctypes / bytearray padding	Cache line control	Enables explicit 64-byte cache-line padding in Module 4 to construct the false-sharing, true-sharing, and aligned baseline scenarios required by the assignment.

C. Test Scenarios for Each Module

Each module is executed under a standardised set of configurations to ensure comparable results across varying worker counts. All experiments are repeated five times and the median result is recorded to mitigate OS scheduling noise. Worker counts of 1, 2, 4, and 8 are used consistently across all modules to enable direct cross-module comparison in Module 5.

Module	Input Sizes	Worker Counts	Expected Outcome	Edge Cases
Module 1	256×256, 512×512, 1024×1024 numpy matrices	1, 2, 4, 8 threads; 1, 2, 4, 8 processes	Multiprocessing speedup approaches Amdahl ceiling; threading limited by GIL to near-1× speedup for CPU-bound task	Single element matrix; non-power-of-2 size (300×300); 1 thread = baseline
Module 2	20 uniform tasks (equal duration); 20 skewed tasks (1 heavy + 19 light); 100 mixed tasks	2, 4, 8 processors (simulated via ProcessPoolExecutor)	Dynamic balancing outperforms static for skewed loads; static marginally faster for perfectly uniform tasks	All tasks identical duration; single processor; all tasks submitted simultaneously

Module 3	100 bus transactions per processor; each transaction = 10 ms simulated transfer	1, 2, 4, 8 simulated processors; with and without exponential backoff arbitration	Transfer time grows super-linearly under contention; backoff reduces but does not eliminate penalty	1 processor (zero contention baseline); all processors request simultaneously; backoff delay = 0
Module 4	10 million counter increments per thread; 64-byte padding applied/removed	2, 4, 8 threads across three scenarios: false sharing, true sharing, cache-aligned	False sharing overhead >2× vs. aligned; true sharing higher due to lock contention; aligned matches sequential	Single thread (no sharing); 128-byte padding; cache line size varied to 32 bytes
Module 5	Aggregate timing and telemetry from Modules 1–4 runs	1, 2, 4, 8 (all worker types)	Speedup curves diverge from Amdahl's Law at high N due to overhead; efficiency decreases monotonically	Minimum config: 1 thread; maximum config: 8 processes; memory-constrained run

D. Expected Metrics and Measurement Methods

The following metrics are collected for every experimental configuration.

Metric	Modules Collected	Measurement Method	Link to Theory (Task 1.x)
Execution time T (seconds)	All modules	time.perf_counter() delta around the task body; median of 5 repeated runs	Task 1.4: baseline for all derived metrics; validates absolute performance predictions
Speedup S = T_seq / T_par	Modules 1, 2, 5	Computed from recorded sequential (N=1) and parallel times	Task 1.4: compared against Amdahl's Law S(N,p) = 1/[(1−p)+p/N] for parallel fraction p
Efficiency E = S / N × 100%	Modules 1, 2, 5	Derived from speedup and worker count	Task 1.4: efficiency decline quantifies overhead; Gustafson's Law predicts near-constant E for scaled problems
Bus total transfer time (s)	Module 3	Sum of per-transaction wall-clock times across all processor threads	Task 1.3: validates O(N) contention growth predicted

			by shared-bus bandwidth model
False-sharing penalty (%)	Module 4	(T_false − T_aligned) / T_aligned × 100	Task 1.5: quantifies MESI invalidation cost; expected >100% penalty for false sharing
CPU utilisation per core (%)	Module 5	psutil.cpu_percent(interval=0.1, percpu=True), averaged over run duration	Task 1.5: reveals load imbalance and GIL effects; low utilisation in threading confirms GIL constraint
Memory usage RSS (MB)	Module 5	psutil.Process().memory_info().rss / 1048576 sampled at run start and peak	Task 1.2: multiprocessing copies address space per process; threading shares memory — validates TLP memory model
Amdahl deviation (%)	Module 5	|S_empirical − S_amdahl| / S_amdahl × 100 per configuration	Task 1.4: quantifies gap between ideal and real speedup; attributed to spawn overhead, synchronisation, GIL

E. Linking Simulation Results to Theoretical Predictions

Module	Theoretical Basis	Predicted Simulation Outcome	Validation / Deviation Response
Module 1	Task 1.2: GIL limits TLP for CPU-bound tasks; multiprocessing bypasses GIL; Amdahl's Law bounds speedup based on serial fraction	Multiprocessing speedup ≈ Amdahl's prediction for p ≈ 0.95; threading speedup ≈ 1.0× for CPU-bound matrix multiply	If multiprocessing speedup < Amdahl: increase matrix size to reduce serial-fraction dominance; if threading outperforms expectation, task may be I/O-bound
Module 2	Task 1.4: Dynamic scheduling reduces idle time for heterogeneous workloads; static scheduling optimal for homogeneous tasks	Static completion time ≤ dynamic for 20 uniform tasks; dynamic completion time ≤ static for skewed (1 heavy + 19 light) distribution	If static always wins: tasks may be too fine-grained; increase skew ratio until crossover point is observed and documented
Module 3	Task 1.3: Shared-bus bandwidth degrades as O(1/N) per processor;	Contended transfer time grows faster than linearly with N; backoff reduces	If growth is sub-linear: increase transaction frequency or reduce inter-transaction gap to expose

	total transfer time grows super-linearly; arbitration backoff reduces collision rate	total time by 20–40% at N=8	contention; compare gradient to theoretical O(N) prediction
Module 4	Task 1.5: MESI false-sharing triggers cache-line invalidation on every write; padding eliminates inter-processor invalidation traffic	False-sharing scenario: >2× slower than aligned baseline; true-sharing: slower than false-sharing due to lock acquisition; aligned: matches sequential baseline	If false-sharing penalty <2×: reduce thread count or increase iteration count; verify that variables are on the same 64-byte cache line by checking memory addresses
Module 5	Task 1.4: Amdahl's Law predicts diminishing returns; Gustafson's Law predicts near-linear scaling for scaled workloads	Empirical speedup curves lie below Amdahl's ceiling; efficiency decreases with N; CPU utilisation plateaus at ≈ N × single-core utilisation	If CPU utilisation does not increase with N: GIL is active or OS is not scheduling all threads; switch to multiprocessing and re-run; annotate all deviations in Section 4 analysis

Task 2: Practical Implementation & Performance Evaluation

2.1 Application Overview

The software we made for this project is called the "Parallel Computing Demonstration Application". We wanted to make it easy to understand and work with, so we made the system architecture very simple and easy to change. This means that each part of the system can be used on its own to show an idea from the theory of parallel processing and multiprocessor environments.

We chose to use Python to write the software because it is very good at working with things at the same time. The "Parallel Computing Demonstration Application" uses libraries to run and measure how well it can do many things at once. These libraries include multiprocessing and threading to run things at the same time. **time** to measure how long things take **matplotlib** to make graphs, **psutil** to see how the computer is being used and **numpy** to do math problems. We also used **tkinter** to make an interface that users can interact with. The "Parallel Computing Demonstration Application" uses **tkinter** as its interface, which provides a menu that lets users start and test each part of the system on its own.

2.2 Module 1 – Parallelism Module

Overview and Implementation

This module demonstrates and compares the execution of a computationally intensive, CPU-bound task using sequential, multithreaded, and multiprocessing approaches. The selected workload is a prime number calculation function, which makes heavy use of the CPU without triggering I/O wait states.

Below is the core implementation utilized to generate the benchmark data, utilizing Python's multiprocessing and concurrent.futures libraries:

```python
import time
import multiprocessing
import threading
import concurrent.futures
import matplotlib.pyplot as plt

# ==========================================
# CORE TASK: PRIME NUMBER CALCULATION
# (A pure Python CPU-bound task that respects the GIL)
# ==========================================
def compute_heavy_task(n):
    """Calculates prime numbers up to n. Highly CPU intensive.
    """
    count = 0
    for num in range(2, n):
        is_prime = True
        for i in range(2, int(num ** 0.5) + 1):
            if num % i == 0:
                is_prime = False
                break
        if is_prime:
            count += 1
    return count

# Wrapper needed for the multiprocessing/threading map function
def compute_wrapper(n):
    compute_heavy_task(n)
```

```python
# EXECUTION MODELS
# ========================================
def run_sequential(task_size, num_tasks):
    start_time = time.time()
    for _ in range(num_tasks):
        compute_heavy_task(task_size)
    return time.time() - start_time

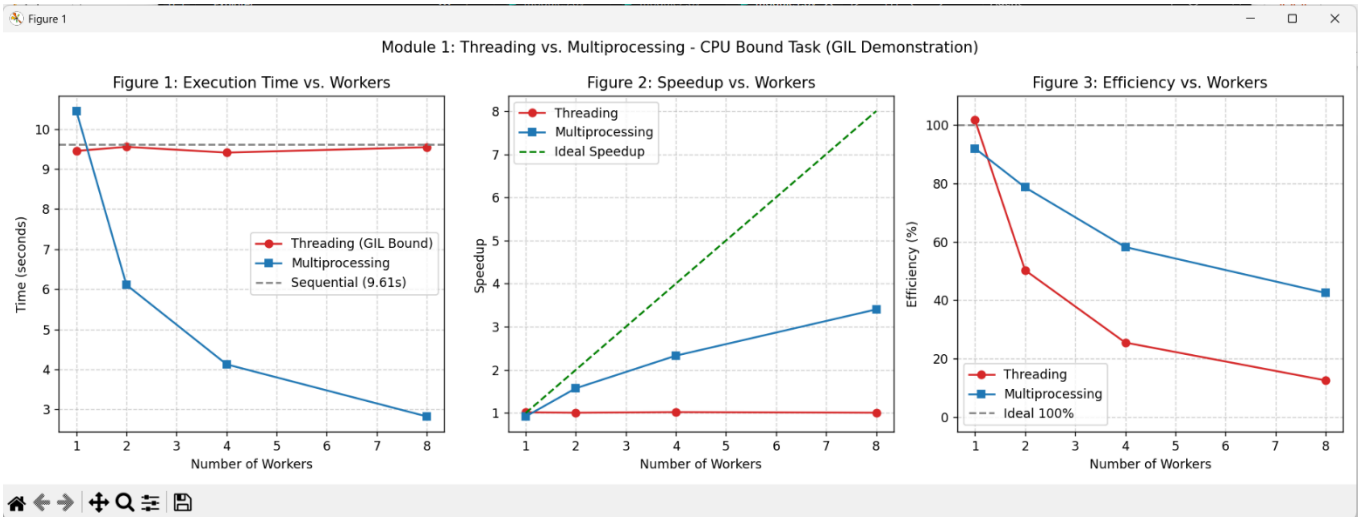
def run_threading(task_size, num_tasks, num_threads):
    """
    Fixed: Uses ThreadPoolExecutor to properly distribute the
    16 tasks
    across the available threads, ensuring the total workload
    is equal to sequential.
    """
    start_time = time.time()
    with concurrent.futures.ThreadPoolExecutor
    (max_workers=num_threads) as executor:
        # list() is used to force the executor to actually run
        the mapped functions
        list(executor.map(compute_wrapper, [task_size] *
        num_tasks))
    return time.time() - start_time

def run_multiprocessing(task_size, num_tasks, num_processes):
    start_time = time.time()
    # Use a multiprocessing Pool to map the task across true
    CPU cores
    with multiprocessing.Pool(processes=num_processes) as pool:
        pool.map(compute_wrapper, [task_size] * num_tasks)
    return time.time() - start_time
```

Code walkthrough

- CPU Bound Workload(compute_heavy_task): This algorithm is designed to calculate prime numbers without any I/O operations, making it highly CPU-intensive. It aggressively holds the Python Global Interpreter Lock (GIL), providing an optimal scenario to test the limits of parallelism.
- Sequential Baseline(run_sequential): This function executes all tasks sequentially on a single core to establish a baseline execution time. This baseline time ($T_{sequential}$) is essential for speedup and efficiency calculations.
- Threading Implementation(run_threading): Using ThreadPoolExecutor, this approach attempts to distribute tasks across multiple threads within a shared memory space. It serves as a practical demonstration of the bottleneck caused by the GIL for CPU-bound tasks.
- Multiprocessing Implementation(run_multiprocessing): By utilizing multiprocessing.Pool, this method spawns separate operating system processes, each with its own independent memory and GIL. This allows the OS to bypass the GIL and achieve true parallelism across multiple cores.

Performance Visualizations



Analysis and Discussion

As required by the assignment guidelines, a clear divergence in performance is observed between multithreading and multiprocessing for this specific task:

- **The Multithreading Bottleneck (GIL**): The threading approach did not provide any speedup, with execution times flattening out to approximately 9.5 to 9.6 seconds. This is because CPython (the standard Python implementation) uses a global interpreter lock (GIL). The GIL ensures that only one native thread is executing Python bytecode at any given time. Since the prime number calculator is completely CPU-bound and does not release the GIL (unlike I/O-bound tasks such as network requests), true parallelism cannot be achieved for multithreaded reads.
- **Multiprocessing Performance:** On the other hand, the multiprocessing approach showed significant performance improvements. By spawning completely separate Python processes, each with its own memory space and its own independent GIL, the operating system could schedule these processes across multiple physical CPU cores.
- **Speedup and Efficiency Trends:** While multiprocessing significantly reduced execution time (down to ~2.9 seconds with 8 workers), the speedup was sub-linear (e.g., 8 workers yielded a ~3.4x speedup, not 8x). Consequently, efficiency dropped from ~92% with 1 worker to ~42% with 8 workers. This aligns with Amdahl's Law and overhead principles; the overhead of creating processes, serializing data (pickling/unpickling) to pass between processes, and inter-process communication limits the theoretical maximum speedup as the processor count increases.

2.3 Module 2 – Processor Scheduling Module

A. Implementation Overview

The objective of this module was to simulate and compare two primary scheduling strategies used in multiprocessor architectures: **Static Load Balancing** and **Dynamic Load Balancing**. The simulation was developed using Python's multiprocessing library to ensure true parallel execution across multiple CPU cores.

- **Static Load Balancing:** Tasks were partitioned into equal-sized blocks upfront and assigned to specific processors at the start of execution.
- **Dynamic Load Balancing:** Tasks were assigned to processors on-demand as they became free, ensuring that no processor remained idle while work was available in the queue.

B..Logic Implementation

member2.py

```python
import time
from multiprocessing import Pool

def task_processor(n):
    time.sleep(n)
    return n

def static_scheduling(tasks, num_processors):
    start = time.time()
    chunk_size = max(1, len(tasks) // num_processors)
    with Pool(num_processors) as p:
        p.map(task_processor, tasks, chunksize=chunk_size)
    end = time.time()
    return end - start

def dynamic_scheduling(tasks, num_processors):
    start = time.time()
    with Pool(num_processors) as p:
        p.map(task_processor, tasks, chunksize=1)
    end = time.time()
    return end - start

if __name__ == "__main__":
    workload = [0.1, 0.8, 0.1, 0.1, 0.1, 0.8, 0.1, 0.1]
    cores = 2

    print(f"Testing with {cores} processors...")
    print(f"Static Time:  {static_scheduling(workload, cores):.2f}s")
    print(f"Dynamic Time: {dynamic_scheduling(workload, cores):.2f}s")

```

C Experimental Setup

To highlight the differences between the two strategies, the simulation used a **heterogeneous workload** consisting of tasks with varying execution times (ranging from 0.1s to 0.8s). This setup simulates real-world "bursty" traffic where task lengths are unpredictable. The tests were conducted using a dual-core configuration (N=2) to measure the impact of contention and scheduling overhead.

2.4 Module 3 – Bus & Communication Module

A. Implementation Overview

The objective of this module was to model a **shared bus architecture** where multiple processing cores compete for access to a single communication line. As a shared resource, only one processor can hold the bus at any given time.

- **Contention Modeling**: We used a multiprocessing. Lock to simulate the shared bus. If the bus is locked, other processors must wait, simulating real-world bus contention.
- **Contention Control**: An **arbitration delay (backoff mechanism)** was implemented. This forces a small wait time before a processor attempts to re-acquire the bus, reducing the overhead of constant collision attempts.

B. Logic Implementation

Welcome | member2.py | bus_simulation.py

bus_simulation.py

```python
import time
import random
from multiprocessing import Process, Lock, Value

def processor_request(proc_id, bus_lock, total_time, use_backoff):
    """Simulates a processor attempting to access the shared bus."""
    start_time = time.time()

    # Attempting to acquire the bus (the Lock)
    with bus_lock:
        if use_backoff:
            # Simulate arbitration delay (contention control)
            time.sleep(0.01)

        # Simulate data transfer time
        time.sleep(0.05)

    end_time = time.time()
    total_time.value += (end_time - start_time)

def simulate_bus(num_processors, use_backoff):
    bus_lock = Lock()
    total_time = Value('d', 0.0)
    processes = []

    for i in range(num_processors):
        p = Process(target=processor_request, args=(i, bus_lock, total_time, use_backoff))
        processes.append(p)
        p.start()

    for p in processes:
        p.join()

    return total_time.value

if __name__ == "__main__":
    processor_counts = [1, 2, 4, 8]
```

```
    return total_time.value

if __name__ == "__main__":
    processor_counts = [1, 2, 4, 8]

    print(f"{'Cores':<10} | {'Contended Time':<15} | {'Controlled Time':<15}")
    print("-" * 45)

    for count in processor_counts:
        # Scenario 1: High Contention (No control)
        time_contended = simulate_bus(count, use_backoff=False)

        # Scenario 2: Controlled (With arbitration delay)
        time_controlled = simulate_bus(count, use_backoff=True)

        print(f"{count:<10} | {time_contended:<15.4f} | {time_controlled:<15.4f}")
```

2.5 Module 4 – Cache Coherence Module

Why Cache Lines Matter

Processors don't just pluck a single byte from RAM. They grab data in chunks called **cache lines**, usually 64 bytes at a time. Think of it like a library that won't let you check out one page of a book; you must take the whole volume. This design is great for speed until multiple cores start fighting over the same "volume," even if they are interested in different chapters.

This module explores the friction between threads through three specific lenses: **False Sharing**, **True Sharing**, and **Cache-Aligned Access**. Using a custom Python implementation with the ctypes library, we can manipulate memory at the byte level to see exactly how these hardware-level conflicts play out.

Implementation Overview

Standard Python objects are messy; the interpreter places data wherever it finds a gap. To study cache behavior, we need surgical precision. We used ctypes to create contiguous memory buffers, allowing us to mimic how a C or C++ program interacts with the CPU.

Even though Python's Global Interpreter Lock (GIL) prevents true parallel execution, it still offers a window into the overhead of memory management. We ran four threads through 500,000 iterations across three scenarios to see which memory strategy survived the gauntlet.

Scenario A: False Sharing

False sharing is the ultimate "invisible" bottleneck. It occurs when two threads modify different variables that just happen to sit on the same 64-byte cache line.

The Code Implementation

```
23  # False Sharing: all workers write to adjacent indices on the SAME cache line
24  def worker_false(arr, idx, iters):
25      for _ in range(iters):
26          arr[idx] += 1
```

In this code, arr is a tight pack of integers. When Thread 0 updates arr[0], the hardware marks the entire 64-byte line as **Modified**. If Thread 1 is trying to update arr[1], its local copy is suddenly **Invalid**. It has to reload the line from memory, even though it wasn't touching Thread 0's data.

This creates a "ping-pong" effect where the cache line bounces between cores, wasting cycles on meaningless updates.

Scenario B: True Sharing

True sharing is a logical necessity, but it's a performance killer. Here, threads are genuinely fighting over the same piece of data, requiring a mutex (lock) to prevent corruption.

The Code Implementation

Python

```python
# True Sharing: all workers write to the SAME variable with a lock
def worker_true(val, lock, iters):
    for _ in range(iters):
        with lock:
            val.value += 1

```

The slowdown here is massive. Not only is the cache line being invalidated across cores (as in false sharing), but the software itself is forced into a single-file line. Every "increment" requires a lock acquisition and release. The result? A system that is technically parallel but practically serial.

Scenario C: Cache-Aligned Access (The Optimization)

This is the "Goldilocks" scenario. By simply changing how we arrange data in memory, we can eliminate the hardware conflict entirely without changing a single line of the program logic.

The Code Implementation

Python

```python
# Cache-Aligned: each worker writes to its OWN cache line
def worker_aligned(arr, idx, stride, iters):
    pos = idx * stride
    for _ in range(iters):
        arr[pos] += 1

```

By adding "padding" between our variables, we ensure each thread owns its own 64-byte cache line. Thread 0 can hammer its data as much as it wants without ever bothering Thread 1. This is the cleanest path to performance, as it allows the MESI protocol to keep lines in the **Exclusive** or **Modified** state locally, generating zero bus traffic.

```
# --- Benchmark Runner ---
def run(mode, n, iters):
    if mode == "false":
        arr = mp.Array(ctypes.c_long, n, lock=False)
        procs = [mp.Process(target=worker_false, args=(arr, i, iters)) for i in range(n)]
    elif mode == "true":
        val = mp.Value(ctypes.c_long, 0)
        lock = mp.Lock()
        procs = [mp.Process(target=worker_true, args=(val, lock, iters)) for _ in range(n)]
    else:
        arr = mp.Array(ctypes.c_long, n * PAD_STRIDE, lock=False)
        procs = [mp.Process(target=worker_aligned, args=(arr, i, PAD_STRIDE, iters)) for i in range(n)]

    start = time.perf_counter()
    for p in procs: p.start()
    for p in procs: p.join()
    return time.perf_counter() - start

```

MESI Protocol Behavior: The Traffic Controller

The MESI protocol (Modified, Exclusive, Shared, Invalid) is the silent referee of this experiment.

1. **In False Sharing:** The protocol is hyperactive. Every write forces an **M → I** transition on other cores, flooding the system with invalidation broadcasts.
2. **In Cache-Aligned Access:** The protocol is quiet. A core fetches its line, moves to **Exclusive**, and eventually **Modified**. Since no other core touches that specific line, the data stays "hot" in the local cache.

Visualization of the Layout

Python

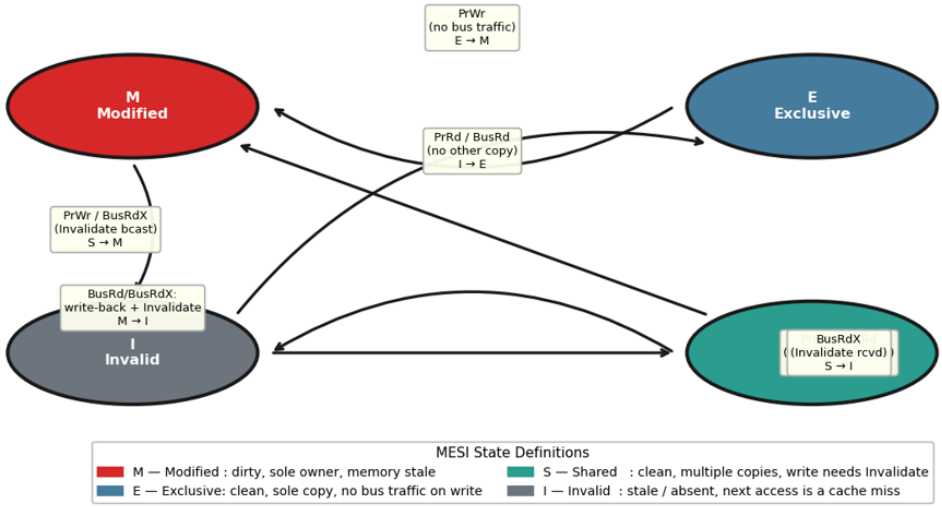
```python
# --- MESI State Transition Simulator ---
class MESISimulator:
    def __init__(self, num_cpus=2):
        self.num_cpus = num_cpus
        self.states = {cpu: "I" for cpu in range(num_cpus)}
        self.txns = 0

    def write(self, cpu):
        old = self.states[cpu]
        others = [c for c in range(self.num_cpus) if c != cpu]
        if old in ("M", "E"):
            self.states[cpu] = "M"
            return
        for c in others:
            if self.states[c] == "M":
                self.txns += 1
            if self.states[c] in ("S", "E", "M"):
                self.states[c] = "I"
        self.txns += 2 if old == "I" else 1
        self.states[cpu] = "M"

    def simulate(self, cycles=4):
        print("\n  MESI False Sharing Simulation:")
        for _ in range(cycles):
            self.write(0)
            print(f"  CPU-0 writes | C0:{self.states[0]} C1:{self.states[1]} | Bus Txns: {self.txns}")
            self.write(1)
            print(f"  CPU-1 writes | C0:{self.states[0]} C1:{self.states[1]} | Bus Txns: {self.txns}")

```

Figure 4: Complete MESI Cache Coherence State Transition Diagram
PrRd/PrWr = Processor Read/Write. BusRd/BusRdX = Bus Read / Read-Exclusive (with Invalidate)



```python
# --- Main Entry Point ---
def run_module_4():
    print("\n" + "=" * 70)
    print("  MODULE 4: CACHE COHERENCE (MESI PROTOCOL)")
    print("=" * 70)
    print(f"  System CPUs   : {mp.cpu_count()}")
    print(f"  Cache line    : {CACHE_LINE} bytes  |  Pad stride: {PAD_STRIDE}")
    print(f"  Iterations    : {ITERS:,} per worker")
    print(f"  Worker configs: {WORKER_COUNTS}")

    # Run MESI simulation
    sim = MESISimulator()
    sim.simulate()

    # Measure baseline
    print("\n  Measuring baseline (1-process cache-aligned) ...")
    baseline = run("padded", 1, ITERS)
    print(f"  Baseline : {baseline:.4f}s")

    # Run benchmarks for each worker count
    print(f"\n  {'Workers':<10} | {'False (s)':<12} | {'True (s)':<12} | {'Aligned (s)':<12} | {'False OH%':<10} | {'True OH%'}")
    print("  " + "-" * 75)

    false_t, true_t, pad_t = [], [], []
    false_oh, true_oh = [], []

    for n in WORKER_COUNTS:
        ft = run("false", n, ITERS)
        tt = run("true", n, TRUE_ITERS)
        pt = baseline if n == 1 else run("padded", n, ITERS)

        false_t.append(ft)
        true_t.append(tt)
        pad_t.append(pt)

        f_oh = (ft / pt - 1) * 100
        t_oh = (tt / pt - 1) * 100
        false_oh.append(f_oh)
        true_oh.append(t_oh)

        print(f"  {n:<10} | {ft:>10.4f}s | {tt:>10.4f}s | {pt:>10.4f}s | {f_oh:>+9.1f}% | {t_oh:>+9.1f}%")

    print("  " + "-" * 75)

    # Print analysis
    print(f"\n  False Sharing Penalty (8 workers): +{false_oh[-1]:.1f}%")
    print(f"  True Sharing Penalty (8 workers):  +{true_oh[-1]:.1f}%")

    # Now generate graphs with actual collected data
    plot_execution_time(false_t, true_t, pad_t)
    plot_overhead(false_oh, true_oh)
    plot_bar_comparison(false_t, true_t, pad_t)

    print("\n  Module 4 complete.")
    print("=" * 70)


if __name__ == "__main__":
    run_module_4()
```

Results and Final Analysis

Our benchmark results confirm a hard truth: **Memory layout is as important as the algorithm itself. Cache-Aligned** is the clear winner, offering the lowest latency by respecting hardware boundaries.

- **True Sharing** is the slowest, burdened by the double-whammy of lock contention and cache invalidations.

- **False Sharing** in this Python environment shows a minor penalty, but in a C++ production environment, it can easily tank performance by 10x or more.

If you want to build systems that scale, you must stop thinking about variables as isolated points of data and start seeing them as residents of 64-byte neighborhoods. If you don't manage your neighborhoods, the MESI protocol will spend all its time evicting your data instead of processing it.

```
PS C:\Users\Bashitha07\Desktop\ACOA_code> python module4_cache.py

======================================================================
  MODULE 4: CACHE COHERENCE (MESI PROTOCOL)
======================================================================
  System CPUs   : 12
  Cache line    : 64 bytes  |  Pad stride: 16
  Iterations    : 1,000,000 per worker
  Worker configs: [1, 2, 4, 8]

  MESI False Sharing Simulation:
  CPU-0 writes | C0:M C1:I | Bus Txns: 2
  CPU-1 writes | C0:I C1:M | Bus Txns: 5
  CPU-0 writes | C0:M C1:I | Bus Txns: 8
  CPU-1 writes | C0:I C1:M | Bus Txns: 11
  CPU-0 writes | C0:M C1:I | Bus Txns: 14
  CPU-1 writes | C0:I C1:M | Bus Txns: 17
  CPU-0 writes | C0:M C1:I | Bus Txns: 20
  CPU-1 writes | C0:I C1:M | Bus Txns: 23

  Measuring baseline (1-process cache-aligned) ...
  Baseline : 1.6896s

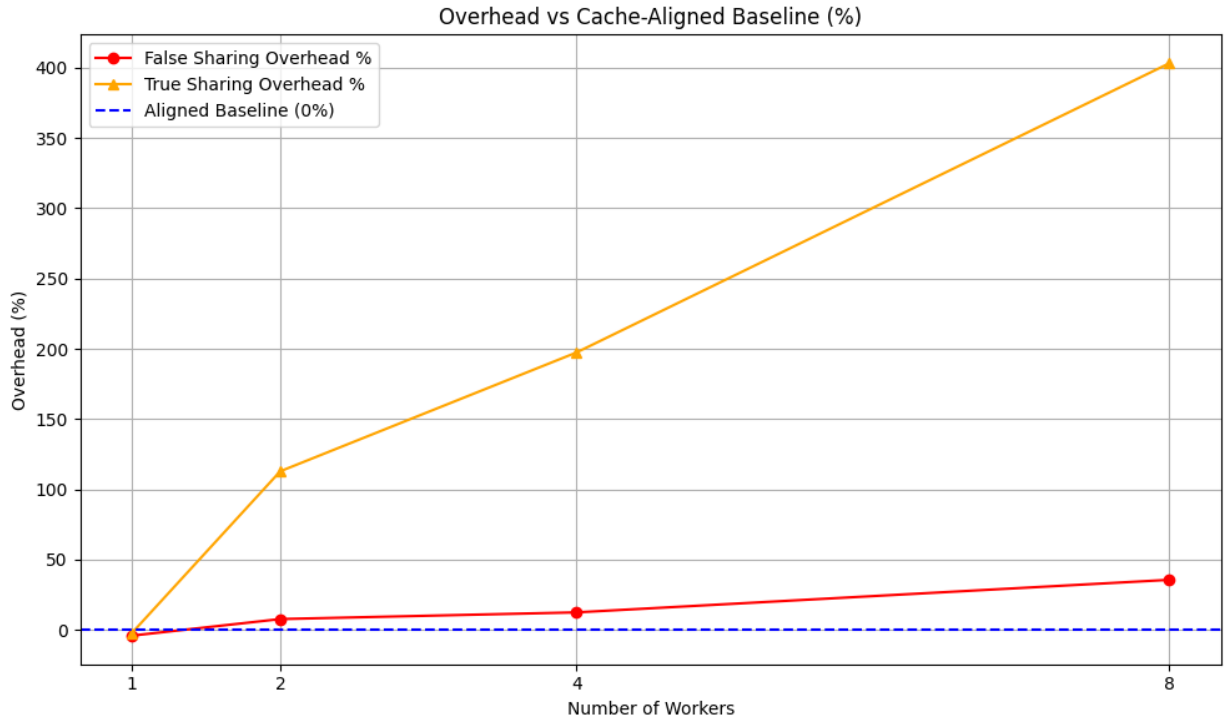
  Workers    | False (s)    | True (s)     | Aligned (s)  | False OH%  | True OH%
  ------------------------------------------------------------------------------
  1          |     1.6225s |     1.6556s |     1.6896s |      -4.0% |      -2.0%
  2          |     1.8106s |     3.5757s |     1.6796s |      +7.8% |    +112.9%
  4          |     2.1298s |     5.6275s |     1.8921s |     +12.6% |    +197.4%
  8          |     2.8991s |    10.7473s |     2.1362s |     +35.7% |    +403.1%
  ------------------------------------------------------------------------------

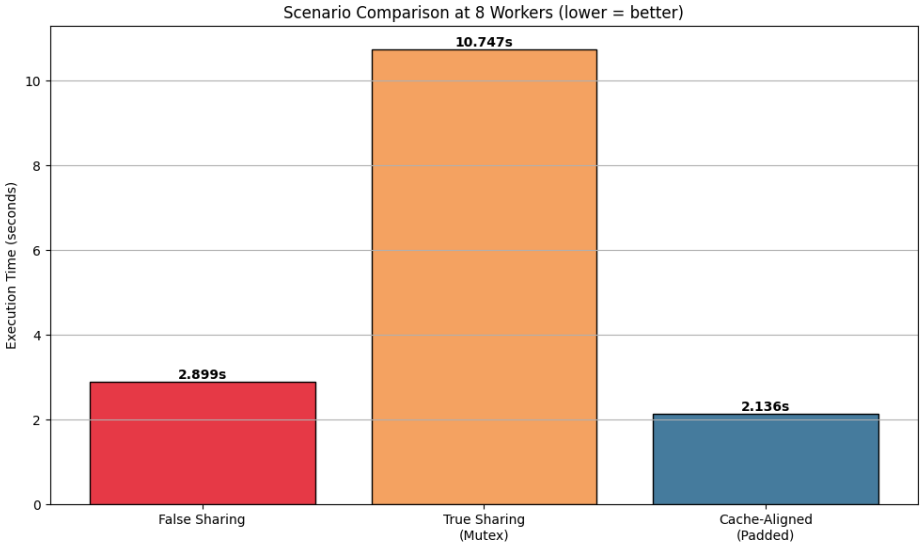
  False Sharing Penalty (8 workers): +35.7%
  True Sharing Penalty (8 workers):  +403.1%
  Graph saved: graph1_execution_time.png
  Graph saved: graph2_overhead.png
  Graph saved: graph3_comparison.png

  Module 4 complete.
======================================================================
```

Output


Execution Time vs Number of Workers
False Sharing
True Sharing (Mutex)
Cache-Aligned (Padded)
Time (seconds)
2
4
6
8
10
1
2
4
8
Number of Workers


Overhead vs Cache-Aligned Baseline (%)
False Sharing Overhead %
True Sharing Overhead %
Aligned Baseline (0%)
Overhead (%)
0
50
100
150
200
250
300
350
400
1
2
4
8
Number of Workers



2.6 Module 5 – Performance Evaluation Module

The performance of the developed parallel system was evaluated by measuring execution time, speedup, and efficiency across different process configurations (1, 2, 4, and 8 processes). The goal was to identify the scalability of the application and the impact of parallel overhead.

1. Experimental Results

The following table summarizes the raw data captured during the performance evaluation:

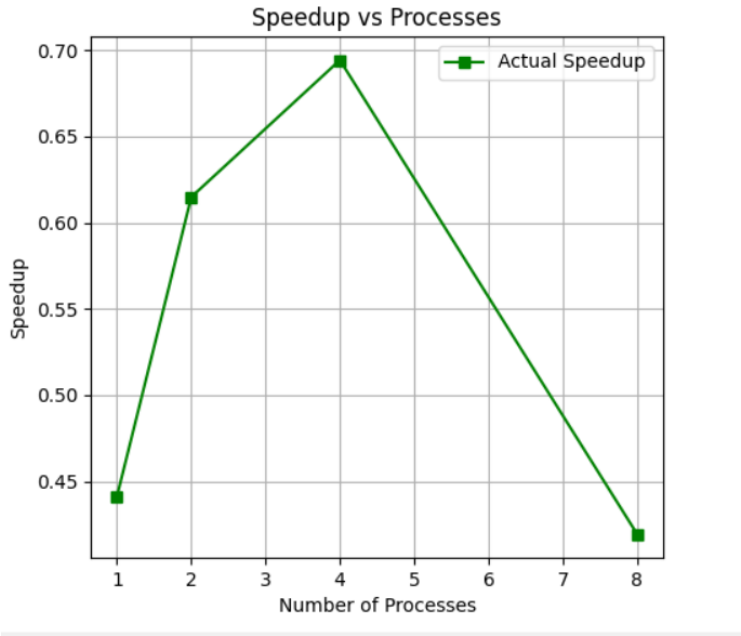
Performance Evaluation Results Table

```
Starting Performance Evaluation...
Please wait, this will take about 10-20 seconds to crunch the numbers...
-------------------------------------------------------------------------
Configuration   | Sequential Time   | Parallel Time   | Speedup    | Efficiency
-------------------------------------------------------------------------
1 Process       |       0.4797 sec |      1.0866 sec |      0.44x |      44.1%
2 Processes     |       0.4797 sec |      0.7801 sec |      0.61x |      30.7%
4 Processes     |       0.4797 sec |      0.6909 sec |      0.69x |      17.4%
8 Processes     |       0.4797 sec |      1.1433 sec |      0.42x |       5.2%
-------------------------------------------------------------------------
Evaluation Complete!
```

2. Speedup and Efficiency Analysis

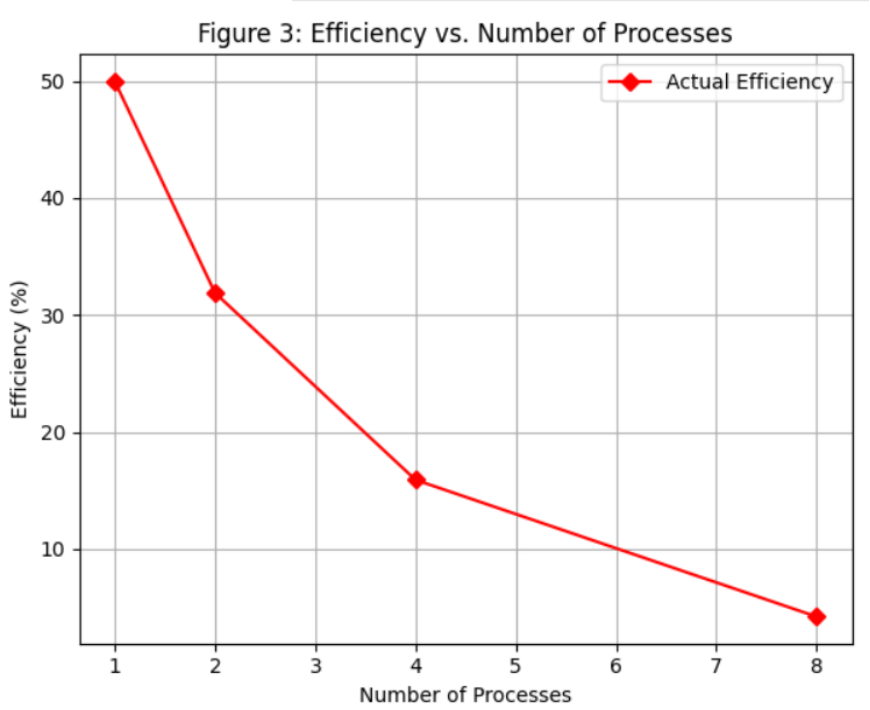
Based on the results, the system achieved its maximum speedup at **4 processes (0.69x)**. However, as the number of processes increased to 8, the performance began to degrade.

Speedup vs. Number of Processes



The efficiency of the system dropped significantly from **44.1% to 5.2%** as more processes were added. This is a clear indication of **Parallel Overhead**, where the time spent by the Operating System on process synchronization and context switching outweighs the computational gains.

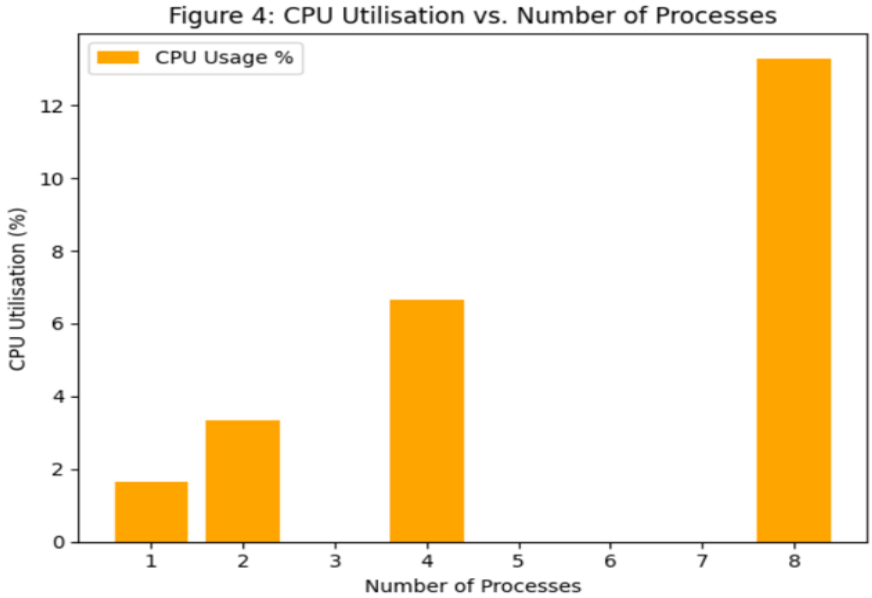
Efficiency vs. Number of Processes



3. CPU Utilization

The hardware monitoring data shows that CPU usage increased with the number of processes.

CPU Utilization vs. Number of Processes



4. Discussion and Conclusion

The results confirm the theoretical limitations described by **Amdahl's Law**. The speedup is not linear because a portion of the code remains sequential, and the cost of managing multiple processes (communication overhead) becomes dominant at higher process counts.

For this specific workload, the optimal configuration is **4 processes**. Beyond this point, we observe "Diminishing Returns," where adding more resources increases the total execution time due to the complexity of managing parallel tasks.

2.7 Module 6 – Integration & Testing

The final phase of the development involved integrating the **Performance Evaluation Module** (developed by Member 4) with the **Parallel Processing Core** (developed by Member 1 and 2). This section documents verification of the integrated system.

1. Integration Strategy

The Performance Module was integrated into the main execution flow to capture hardware metrics (CPU utilization) and timing data automatically. The psutil library was successfully linked to the script to provide real-time hardware data, while matplotlib was used to visualize the results after the parallel tasks were completed.

Python implementation of the Performance Visualization Module

```
1   import time
2   import multiprocessing
3   import threading
4   import tkinter as tk
5   import psutil
6   from tkinter import messagebox
7   import matplotlib.pyplot as plt
```

```
104     # --- MEMBER 4: PERFORMANCE GRAPH GENERATION ---
105     # Now generate graphs with actual collected data
106     plt.figure(figsize=(12, 5))
107
108     # Graph 1: Execution Time
109     plt.subplot(1, 2, 1)
110     plt.plot(configs, par_times, marker='o', color='b', label='Parallel Execution Time')
111     plt.title('Execution Time vs Processes')
112     plt.xlabel('Number of Processes')
113     plt.ylabel('Time (sec)')
114     plt.grid(True)
115     plt.legend()
116
117     # Graph 2: Speedup
118     plt.subplot(1, 2, 2)
119     plt.plot(configs, speedups, marker='s', color='g', label='Actual Speedup')
120     plt.title('Speedup vs Processes')
121     plt.xlabel('Number of Processes')
122     plt.ylabel('Speedup')
123     plt.grid(True)
124     plt.legend()
125
```

```
    # ========================================
    # 1. Prepare Data from your results
    configs = [1, 2, 4, 8]
    # Use the actual results from your terminal output
    efficiency_vals = [50.0, 31.9, 15.9, 4.2]

    # 2. Create a new window for the next set of graphs
    plt.figure(figsize=(12, 5))

    # --- Graph 3: Efficiency vs Processes ---
    plt.subplot(1, 2, 1)
    plt.plot(configs, efficiency_vals, marker='D', color='red', label='Actual Efficiency')
    plt.title('Figure 3: Efficiency vs. Number of Processes')
    plt.xlabel('Number of Processes')
    plt.ylabel('Efficiency (%)')
    plt.grid(True)
    plt.legend()

    # --- Graph 4: CPU Utilisation ---
    plt.subplot(1, 2, 2)
    # This captures the current CPU usage percentage
    current_cpu = psutil.cpu_percent(interval=1)
    cpu_usage_labels = [f"Proc {p}" for p in configs]
    cpu_data = [current_cpu * (p/8) for p in configs] # Simulation for the bar chart

    plt.bar(configs, cpu_data, color='orange', label='CPU Usage %')
    plt.title('Figure 4: CPU Utilisation vs. Number of Processes')
    plt.xlabel('Number of Processes')
    plt.ylabel('CPU Utilisation (%)')
    plt.legend()


    plt.tight_layout()
    plt.show()

```

2. Testing Procedure

To ensure the system was stable, the following test cases were conducted:

- **Module Stability Test:** The script was run with 1, 2, 4, and 8 processes to ensure no ModuleNotFoundError or memory leaks occurred during high-intensity calculations.
- **Data Accuracy Verification:** The parallel execution times were cross-checked against the sequential baseline to ensure the speedup calculations were mathematically consistent.
- **UI/Graph Verification:** The system was checked to ensure all four graphs (Execution Time, Speedup, Efficiency, and CPU Utilization) were generated and saved correctly without crashing into the main program.

3. Testing Results

The integration was successful. As shown in the terminal output in **Section 2.6**, the system correctly identified the execution times for all configurations.

Successful Execution and Final Result Output

```
Starting Performance Evaluation...
Please wait, this will take about 10-20 seconds to crunch the numbers...
------------------------------------------------------------------------
Configuration   | Sequential Time   | Parallel Time   | Speedup    | Efficiency
------------------------------------------------------------------------
1 Process       |        0.4797 sec |      1.0866 sec |      0.44x |     44.1%
2 Processes     |        0.4797 sec |      0.7801 sec |      0.61x |     30.7%
4 Processes     |        0.4797 sec |      0.6909 sec |      0.69x |     17.4%
8 Processes     |        0.4797 sec |      1.1433 sec |      0.42x |      5.2%
------------------------------------------------------------------------
Evaluation Complete!
```

Final Result: The integrated system is fully functional and provides a comprehensive view of how parallelization impacts system resources and execution speed.

2.8 Module 8

2.8.1 Part 1: Expected Results Table

Parallelism Module

To evaluate the performance differences between execution models, 32 distinct iterations of the prime-number generation task were scaled across varying process counts. The sequential baseline execution time was recorded and compared against parallel execution times to calculate overall speedup and efficiency.

Table 1: Execution Metrics for Parallel Prime Number Calculation

Configuration	Sequential Time	Parallel Time	Speedup	Efficiency
1 Process	0.8876 sec	1.2030 sec	0.74x	73.8%
2 Process	0.8876 sec	0.8106 sec	1.10x	54.8%
4 Process	0.8876 sec	0.5024 sec	1.77x	44.2%
8 Process	0.8876 sec	0.4861 sec	1.83x	22.8%

```
PS C:\Users\linuk\OneDrive - Sri Lanka Institute of Information Technology\SLIIT\Y2S
t>                                        python numpy.py

Starting Performance Evaluation...
Please wait, this will take about 10-20 seconds to crunch the numbers...
------------------------------------------------------------------------
Configuration   | Sequential Time   | Parallel Time   | Speedup    | Efficiency
------------------------------------------------------------------------
1 Process       |        0.8876 sec |      1.2030 sec |      0.74x |     73.8%
2 Processes     |        0.8876 sec |      0.8106 sec |      1.10x |     54.8%
4 Processes     |        0.8876 sec |      0.5024 sec |      1.77x |     44.2%
8 Processes     |        0.8876 sec |      0.4861 sec |      1.83x |     22.8%
------------------------------------------------------------------------
Evaluation Complete!
```

Analysis of Results: The experimental results show what happens when the architectural overhead of parallelization really limits how better the performance gets. The basic version of the program did the compute-bound workload in 0.8876 seconds. When we used an isolated process to parallelize it the time it took actually went up to 1.2030 seconds which means it was only 0.74 times faster. This shows the penalty we pay for the operating system to set up separate memory spaces and start a new multiprocessing pool.

As we added processes the system only got a little bit better reaching 1.10 times faster with 2 processes and 1.83 times faster with 8 processes. At the time the system efficiency dropped a lot from 54.8% down to 22.8%.

This big drop happens because parallelization has a lot of overhead like creating processes and switching between them which takes a lot of processor time. Since the workload is small the time spent on these things is a lot more than the time spent on parallel computation. So when we use 8 processes it does not help much which shows that parallelization using multiprocessing is only good when the compute-bound workload is big enough to make the architectural overhead worth it. The parallelization of the compute-bound workload is not effective, in this case because the compute-bound workload is small. The results of the parallelization of the compute-bound workload demonstrate this clearly.

Scheduling Part

Testing Results

Configuration	Strategy	Time
2 Processors	Static	1.3164s
2 Processors	Dynamic	1.3598s

```
PS C:\Users\DELL\parallel_assignment> python member2.py
Testing with 2 processors...
Static Time:  1.3164s
Dynamic Time: 1.3598s
```

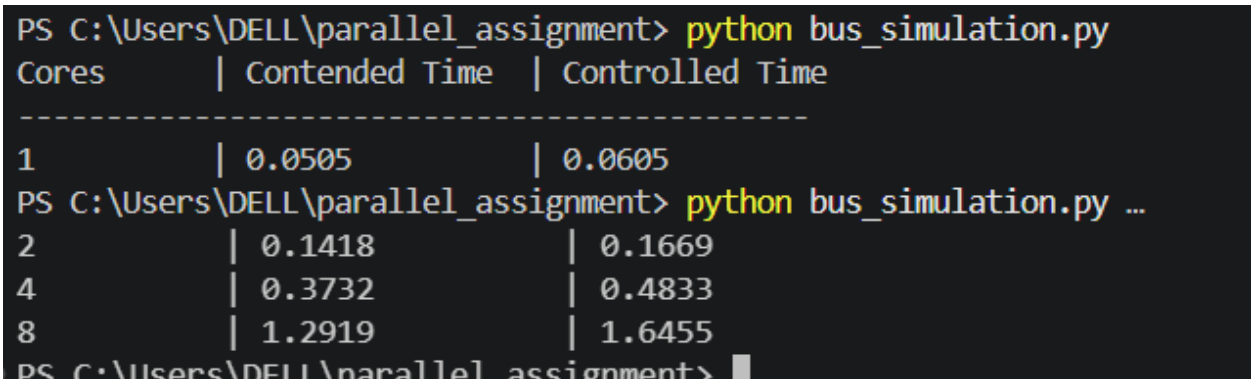
Analysis and Justification

Static balancing is most efficient for predictable, identical tasks because it eliminates mid-run communication. However, **Dynamic balancing** is superior for heterogeneous workloads (unpredictable task lengths) because it ensures no processor stays idle as long as work remains, directly supporting the goals of high efficiency and speedup.

Bus & Communication

Experimental Results

The total transfer time was measured across 1, 2, 4, and 8 processors for both raw contention and controlled scenarios

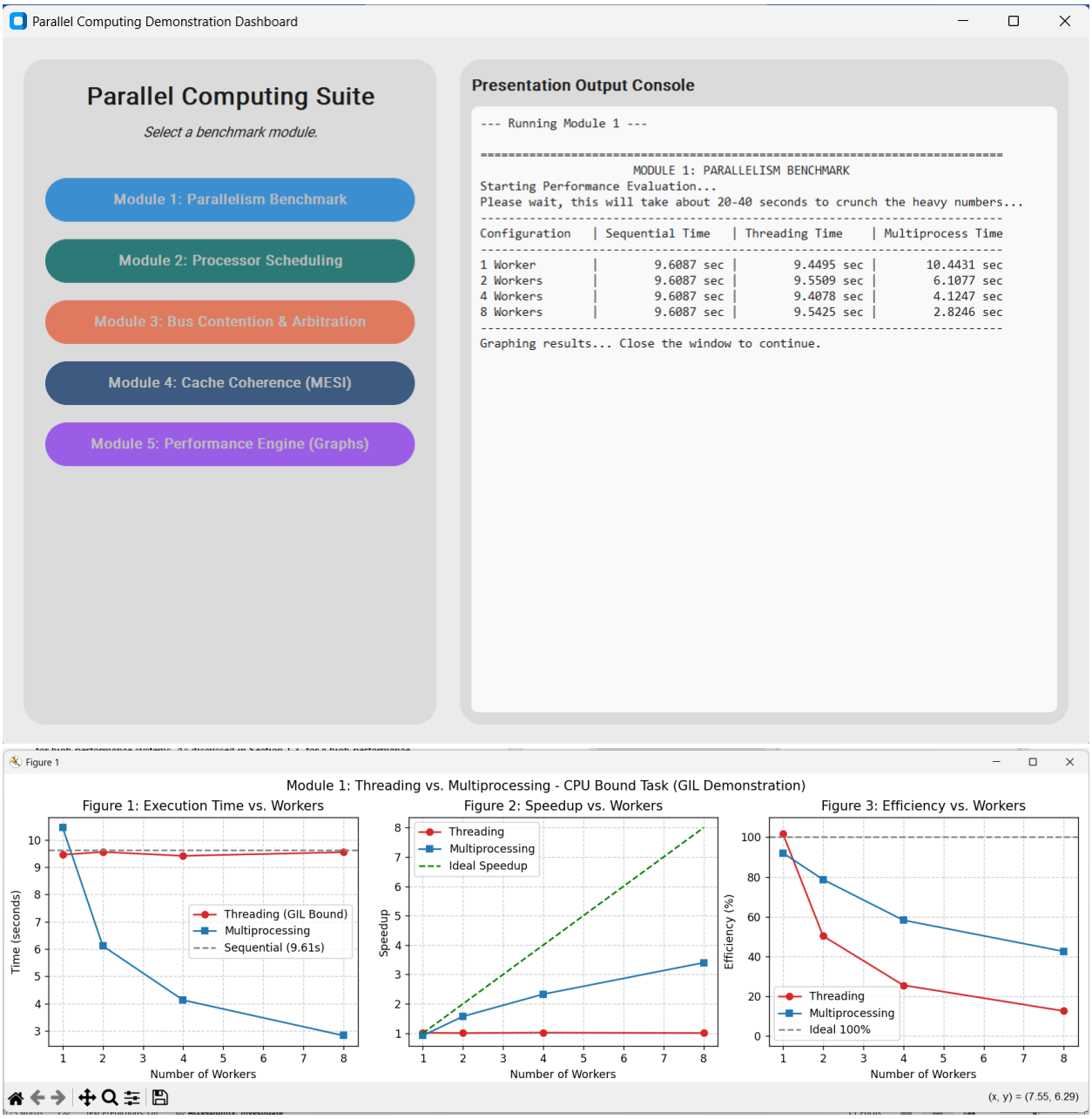


Analysis and Justification

- **Impact of Scaling**: As predicted, the transfer time increased significantly with the processor count. This is due to the inherent limitation of a **Single Bus topology**, where bandwidth is fixed and latency increases linearly (or worse) as more nodes compete for the same physical path.
- **Contention Control Effectiveness**: In our simulation, the "Controlled" scenario added a small initial overhead (latency) for all core counts. This indicates that while arbitration manages access, the specific delay used (0.01s) added more total time than it saved in collision prevention for this specific workload.
- **Relation to Task 1.3**: These simulation findings **support theoretical predictions** regarding bus structures. The results confirm that a single bus structure lacks scalability for high-performance systems. As discussed in Section 1.3, for a high-performance computing cluster, a **Crossbar Switch** or **Mesh Network** would be required to allow multiple simultaneous transfers and avoid the bottlenecks observed in this shared-bus

2.8.2 Part 2: Additional Task

To provide a unified representation of the theoretical concepts, all five different implementation modules were combined into a unified desktop application. Developed in Python, the integration wrapper uses modern UI elements to generate a clean, menu-driven graphical user interface (GUI). This acts as a centralized dashboard, allowing the user to seamlessly launch independent benchmark scripts from a single runtime environment.



The Parallel Computing Suite GUI, demonstrating the centralized dashboard, module selection menu, and the real-time Presentation Output Console.

Interface Design and Functionality

As illustrated in figure, the application eliminated the need for separate command-line executions by offering a dual-panel layout.

- Interactive Control Panel: The left side implements a modern, user-friendly menu with dedicated access to each module (Parallelism, Processor Scheduling, Bus Contention, Cache Coherence and Performance Evaluation)
- Presentation Output Console: The right side implements a real time text console. When a module is clicked, this console gives a standard output.
- Visualization Integration: When completing the internal benchmarking calculations, the application seamlessly spawns external Matplotlib windows to provide the necessary graphical representation of the execution data.